

Visual Composition

Project: Verified Study Buddy

1. Your mission

Your learning target

I can explain the mechanism and terminology of image generation and provenance.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Structured Output and Schemas. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 11: Visual Composition

Subject, setting, framing, relationships, and constraints shape an image request.



CREATOR MISSION

Diagnose and repair three ambiguous visual prompts.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Generation failures and verification. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

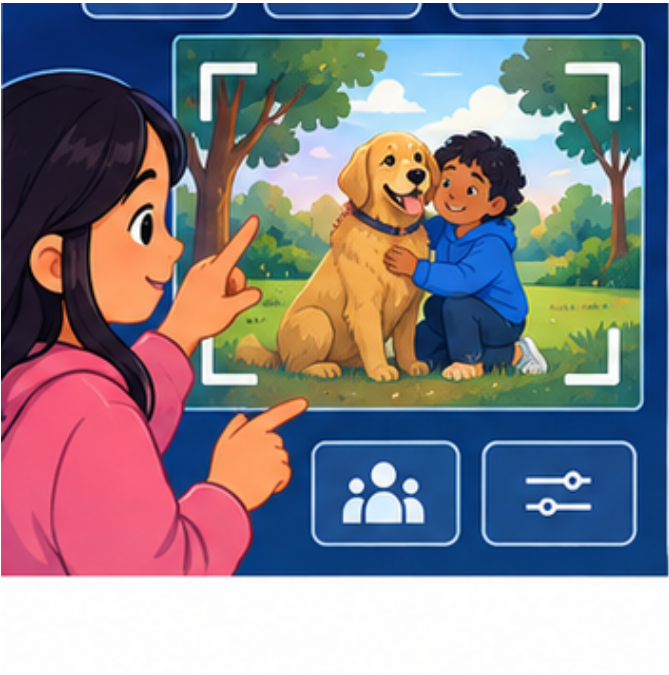
This week's end goal

By the end, I can check object counts and spatial relations as separate visual requirements.

2. Notice the evidence

PICTURE STORY

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use visual composition to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

An image generator constructs an output from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

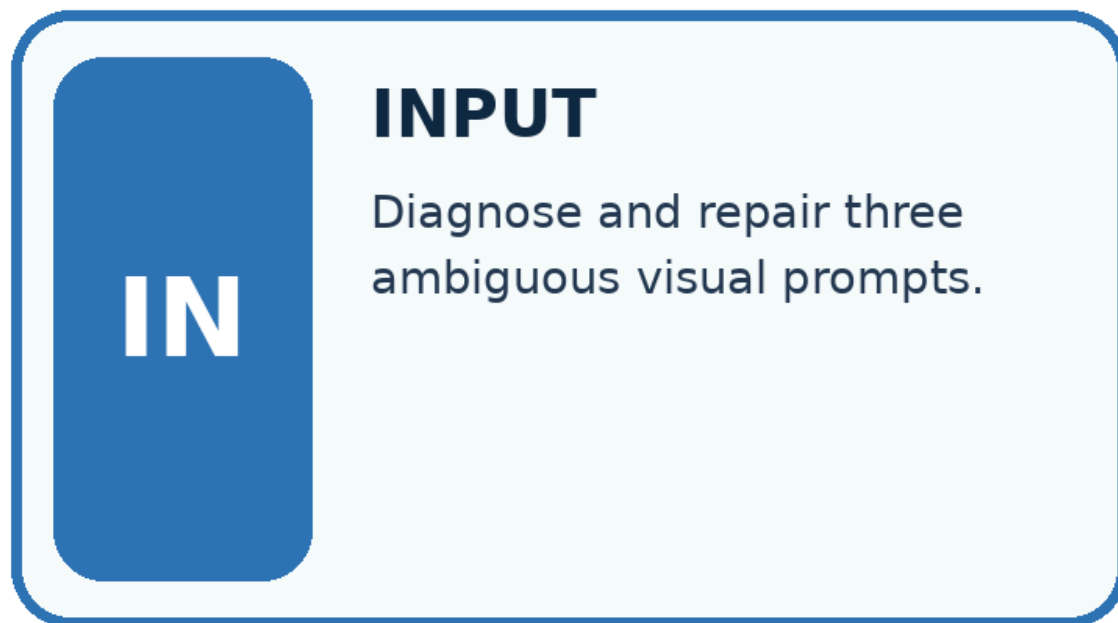
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Subject, setting, framing, relationships, and constraints shape an image request.



OUT

EVIDENCE

Diagnose and repair three ambiguous visual prompts.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery. Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Image generation and provenance — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

Check or calculation: Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.

Conclusion and limit: It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL
Expected learner request



AMBIGUOUS
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Prediction: Position now passes; arrow direction still fails.

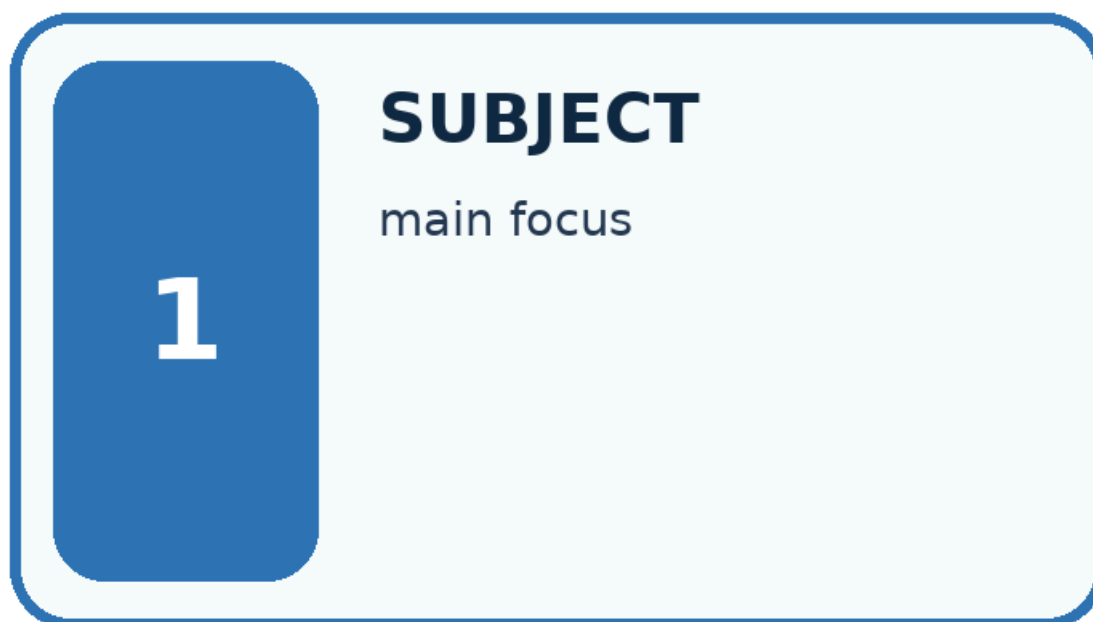
Reason: Changing object placement does not automatically reverse an arrow's semantic source and destination.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

SETTING

environment around the subject

3

FRAMING

viewpoint and crop

4

RELATIONSHIP

how objects or people
connect

Word · My own example · Not this / boundary

Word: subject

My own example: The main person, object or topic represented in an image or piece of content.

Not this / boundary: _____

Word: setting

My own example: A chosen configuration value that affects how a system operates.

Not this / boundary: _____

Word: framing

My own example: The way a question or situation is presented, which can influence interpretation.

Not this / boundary: _____

Word: relationship

My own example: A connection between quantities, events or concepts; an association need not be causal.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Provenance

Meaning: Recorded origin and transformation history

Word: Visual criterion

Meaning: A stated requirement checked against an image

Word: Authenticity claim

Meaning: A claim about the source or reality of depicted content

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 11 • WORKED EVIDENCE

- 1 Prompt names a robot and child but not their relationship.
- 2 Model may place them far apart or reverse who teaches whom.
- 3 Add action, viewpoint, spacing, and must-show constraints.
- 4 Evaluate whether composition communicates the learning idea without misleading detail.

1. Prompt names a robot and child but not their relationship.
2. Model may place them far apart or reverse who teaches whom.
3. Add action, viewpoint, spacing, and must-show constraints.

4. Evaluate whether composition communicates the learning idea without misleading detail.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Visual Composition

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

The facts we will use

Brief: exactly two labelled bottles left of one beaker; one arrow from bottles toward beaker.

Supplied drawing description: two labelled bottles are right of a beaker; an arrow points from beaker toward bottles.

Visual Composition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count objects

Two bottles and one beaker satisfy the count requirement.



2. Check position

Bottles on the right fail the required left-of relation.



3. Check direction

Arrow from beaker to bottles reverses the required flow.



4. Repair brief

Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

CHANGE ONE THING

Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Count objects

What happens and why: Two bottles and one beaker satisfy the count requirement.

Step: 2. Check position

What happens and why: Bottles on the right fail the required left-of relation.

Step: 3. Check direction

What happens and why: Arrow from beaker to bottles reverses the required flow.

Step: 4. Repair brief

What happens and why: Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Having every requested object is not the same as conveying the requested relationship. Position and arrow direction can change a scientific explanation while object counts remain correct. Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

What this example does not establish

A compliant layout does not by itself verify the chemistry or procedure represented; content needs source checking.

Read the labelled picture: Which checks pass after that single change, and what still needs repair?

Change one thing: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: More visual adjectives always create a clearer composition. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and provenance”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Image generation and provenance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which checks pass after that single change, and what still needs repair?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Repair three prompts by changing one missing composition element; sketch expected framing before generation and compare.

Repair three prompts by changing one missing composition element; sketch expected framing before generation and compare.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

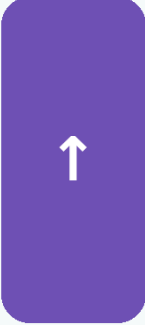
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY
1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY
Create age-fit explanatory visuals with labeled purpose and no invented documentary evidence.

THIS WEEK'S CHECKPOINT Create age-fit explanatory visuals with labeled purpose and no invented documentary evidence.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

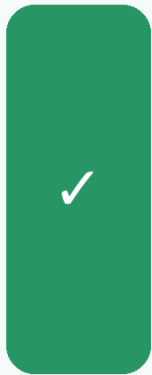
Create Study Buddy diagram rubrics covering counts, readable labels, containment, order and meaningful arrow direction.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Diagnose and repair three ambiguous visual prompts.

CHALLENGE Diagnose and repair three ambiguous visual prompts.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 11 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Brief: large tree left, small bicycle right, open sky above. Output puts bicycle left.

The question

Create a visual checklist.

Worked explanation

Check subjects, relative size, positions and sky separately. The position violates composition even if subjects exist.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Map the brief to regions

Evidence or stage · Value or result

Evidence or stage: Upper region

Value or result: Open sky

Evidence or stage: Left foreground

Value or result: Large tree

Evidence or stage: Right foreground

Value or result: Small bicycle

Brief: large tree left, small bicycle right, open sky above. Output puts bicycle left.

Create a visual checklist.

Check subjects, relative size, positions and sky separately. The position violates composition even if subjects exist.

14. Try a new case independently

Independent case: Changed independent case

A supplied drawing description says three stars are above a triangle, while a circle crosses the edge of a rectangle. The brief requires three stars above the triangle and the circle fully inside the rectangle.

Evaluate this description.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 11 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A poster puts pale yellow labels on white and hides the main object behind decorations. Suggest two revisions and how to check them.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create age-fit explanatory visuals with labeled purpose and no invented documentary evidence.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

An image brief requires two trees, a bench between them and the sun upper-right. A result has all objects but the bench is left of both trees. Write a spatial test that detects this.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a

model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of image generation and provenance

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of image generation and provenance

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or

assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy requests a science-poster layout, then audits a supplied drawing description.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output.

- 1. Check position
- 2. Repair brief
- 3. Count objects
- 4. Check direction

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Repair brief Specify bottles-left, beaker-right, arrow left-to-right; verify the revised output. Having every requested object is not the same as conveying the requested relationship. Position and arrow direction can change a scientific explanation while object counts remain correct. Audit the provided description honestly: no actual image was supplied, so this evaluation concerns its stated layout rather than a claimed visual inspection.

Practice 2: explain the deciding evidence

Changed-case practice: Move bottles left of the beaker but keep the arrow pointing from beaker toward bottles. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future input.

2. Position now passes; arrow direction still fails.

3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Position now passes; arrow direction still fails. Changing object placement does not automatically reverse an arrow's semantic source and destination.

Repair a missing building block

Review Image Generation Models (ai-3/10). Goal: Explain the mechanism and terminology of image generation and provenance. Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Structured Output and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?. Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.